

Mitigating Thermal Crosstalk Using Integrated Thermometry

Sai Kanth Dacha,^{1,*} Karl J. McNulty,² Tiffany B. Nguyen,² Yun Zhao,^{1,3} Michal Lipson,^{2,1} and Alexander L. Gaeta^{1,2}

¹Department of Applied Physics and Applied Mathematics, Columbia University, New York, NY 10027, USA

²Department of Electrical Engineering, Columbia University, New York, NY 10027, USA

³Department of Applied Physics and Ginzton Laboratory, Stanford University, Stanford, CA 94305, USA

*saikanth.dacha@columbia.edu

Abstract: Thermal crosstalk is a major obstacle to operation of photonic integrated circuits with multiple tunable elements. We demonstrate a scalable approach using integrated thermometry that reduces crosstalk between closely-spaced photonic components by over thirty-fold.

1. Introduction

Resistive heating is the most widely used method for tuning photonic integrated devices. It relies on the sensitivity of waveguide length or refractive index to temperature. In a typical thermal tuning scheme, electrical power is applied to a metallic or doped-semiconductor resistive element that is proximal to the photonic device. Photonic chips with multiple components consist of multiple such heating elements. Diffusion of heat within the photonic chip causes thermal crosstalk, in which heat generated by one heating element elevates the temperature of other photonic components. Thermal crosstalk is a major bottleneck for applications such as data communications transceivers [1], neuromorphic computing [2], and nanophotonic phased arrays [3].

Passive nanofabricated structures such as trenches [4] and undercuts [5] have been shown to partially mitigate thermal crosstalk. However, they significantly increase fabrication costs and complexity. Control algorithms based on computational methods such as thermal eigenmode decomposition [6] and subarray design [7] have been proposed. However, such computationally intensive methods are ill-suited for large-scale photonic circuits.

In this work, we present a highly general approach for active suppression of thermal crosstalk using integrated resistance thermometry [8]. Our approach removes the requirement for complex nanofabrication, requires no photodetection, is compatible with a wide variety of material platforms, and is fully foundry compatible. In the proof of concept reported here, we demonstrate an over-thirty-fold crosstalk reduction in a photonic device consisting of a high-quality-factor (high- Q) silicon-nitride (SiN) microresonator and two add-drop filters on a silicon substrate.

2. Thermal Design

Figure 1a depicts the simulated (COMSOL 6.1) steady-state temperature for a heat power $P_{\text{pert.}} = 100$ mW applied to the platinum microheater above a SiN add-drop filter ($R_0 = 30$ μm). Figure 1b shows a line cut of the temperature distribution alongside the Si substrate conduction limit for $z = 0$, i.e., at the same vertical level as SiN. The two curves converge for radial distances from the microheater as short as ≈ 20 μm , indicating that thermal conduction at larger length scales occurs primarily via the Si substrate, as depicted in the inset. This insight forms the basis for the crosstalk suppression scheme described below. Device cross-section and fabrication are described in ref. [8].

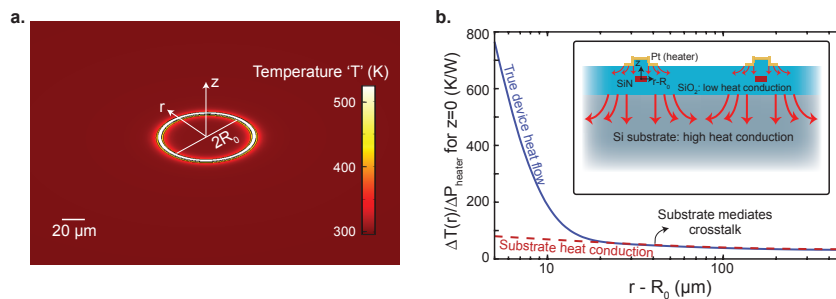


Fig. 1. **a.** Numerically simulated steady-state temperature profile for a thermally-tunable SiN add-drop filter for a microheater power of 100 mW. **b.** Line cut of steady-state temperature distribution, plotted as the thermal-power-to-temperature transduction. Dashed line indicates the Si substrate conduction limit. Inset graphic indicates heatflow pathways in the photonic chip.

3. Mitigating Thermal Crosstalk: Experimental Results

Figure 2a depicts a simplified experimental schematic and microscope picture of the photonic chip fabricated. The crosstalk cancellation scheme functions as follows: a pair of thin-film platinum resistors envelops the resonators of interest, and one acts as a resistance thermometer and the other as a feedback heater. As the power applied to the heater of ring 2 is varied, the temperature change measured by the thermometer is expected to be approximately equal to that experienced by rings 1 and 3 due to the substrate-driven heat flow and the weak range dependence of thermal transduction for length-scales larger than $\approx 20 \mu\text{m}$ (Fig. 1b). The feedback heater is roughly equidistant from the thermometer and the rings, and imparts approximately equal heat to both.

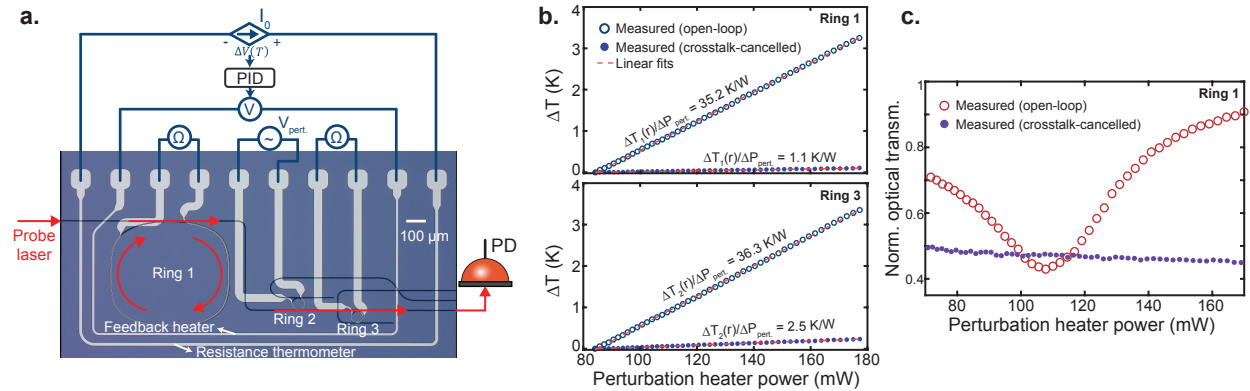


Fig. 2. **a.** Experimental schematic for crosstalk cancellation. The probe laser is initially resonant with ring 1. **b.** Temperature changes of ring 1 and 3, measured using resistance thermometry, for varying ring 2 heater powers, in open-loop and closed-loop configurations. **c.** Normalized transmission from ring 1 for varying ring 2 heater powers, in open-loop and closed-loop configurations.

Figure 2b shows the temperature changes at rings 1 and 3 measured using resistance thermometry, in open-loop and closed-loop (i.e., crosstalk-cancelled) configurations, for a slow sweep of the perturbation power applied to the heater of ring 2. Rings 1 and 3 experience thermal crosstalk from ring 2 heater of 35.2 K/W and 36.3 K/W respectively, quantified here as the slope of the linear fit of the measured temperature change versus heater power, $\Delta T_{1(3)}/\Delta P_{\text{pert}}$. When the feedback loop based on the thermometer's temperature is closed, the measured crosstalk at each ring drops significantly to 1.1 K/W and 2.5 K/W, that is, crosstalk suppression of $32\times$ and $14.5\times$, respectively. The difference in suppression factor may stem from the difference in lateral spacings between ring pairs 1-2 and 2-3. Figure 2c shows the change in normalized optical transmission versus perturbation heater power. The probe laser is manually tuned to be in resonance with ring 1 for this measurement. Without crosstalk suppression, thermal crosstalk from scanning the perturbation heater power by 100 mW alone is sufficient to tune across a high- Q resonance of ring 1. When crosstalk is actively cancelled out, the probe laser remains stably in resonance with ring 1 and exhibits $< 10\%$ variation in transmission over the same range of perturbation heater power.

4. Conclusion

In conclusion, we report a highly general approach to solving thermal crosstalk using resistance thermometry. By employing a straightforward design that leverages the key role played by the Si substrate in chip-wide thermal conduction, we demonstrate an over-thirty-fold crosstalk reduction. Our technique is readily applicable to a variety of material platforms and to linear and nonlinear photonic devices, and can lead to dramatic simplification of control algorithms for complex photonic circuits.

References

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